Chapter test

18 Movement in humans

Name: _____ ( ) Class: _____ Date: _____

Total marks: 40 marks
Time allowed: 40 minutes
Answer ALL questions.

Score: / 40

A. Concept checking (10 marks, 1 mark each)

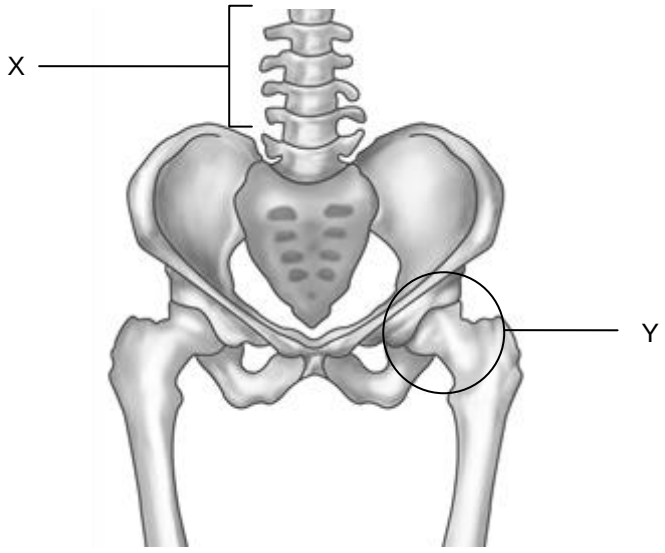
1. Which of the following comparisons between bones and cartilage are ***incorrect***?

	Bones	***Cartilage***
A.	living	non-living
B.	less elastic	more elastic
C.	contain more minerals	contain less minerals
D.	with blood supply	without blood supply

2. Which of the following does ***not*** contain cartilage?

A. ear pinna
B. bronchus
C. rib cage
D. penis

Directions: Questions 3 and 4 refer to the diagram below, which shows a pelvic girdle and its associated bones.



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3. Which of the following statements about X is ***incorrect***?

   A. It provides surfaces for muscle attachment.
   B. It protects the spinal cord.
   C. It is a site of breakdown of old red blood cells.
   D. It consists of bones and cartilage. ☐

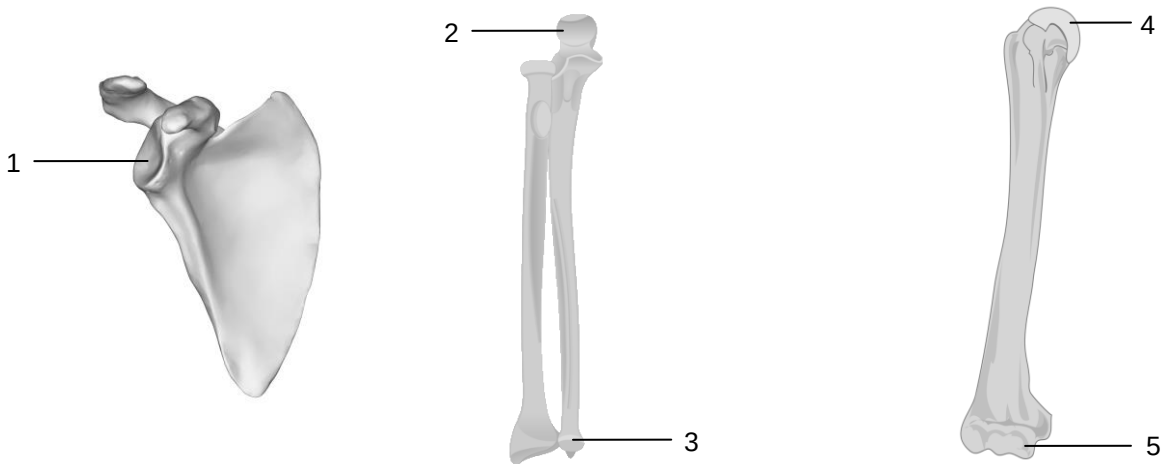
4. Y is a joint that allows movement in

   A. one plane only.
   B. two planes only.
   C. three planes only.
   D. many planes. ☐

5. In a pregnant woman, which of the following may be supplied by her bones to the foetus?

   A. blood cells
   B. iron
   C. calcium
   D. antibodies ☐

★ 6. The diagram below shows the shoulder bone and some bones from the arm.



Between which parts is a ball-and-socket joint formed?

   A. 1 and 2
   B. 1 and 4
   C. 2 and 4
   D. 2 and 5 ☐

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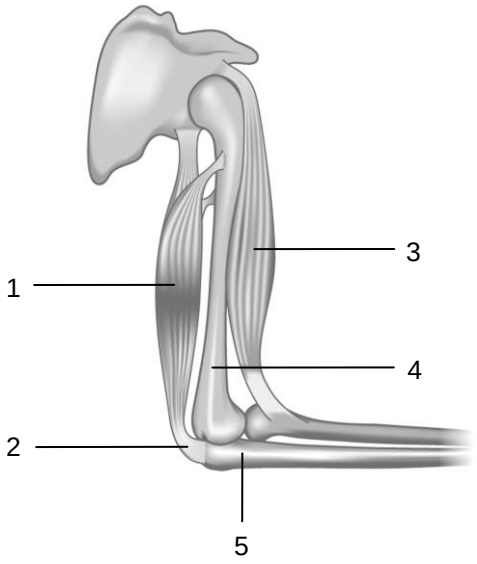
7. Which of the following statements is ***incorrect***?

   A. A skeletal muscle becomes bigger when it contracts.
   B. Skeletal muscles are attached to bones by tendons.
   C. Skeletal muscles can only pull bones but cannot push.
   D. Skeletal muscles are voluntary muscles. ☐

★ 8. Which of the following liquids, when added to some fresh muscle fibres, would result in the most immediate contraction of the muscle fibres?

   A. distilled water
   B. 1% glucose solution
   C. 1% sucrose solution
   D. 1% ATP solution ☐

9. The diagram below shows some structures associated with the elbow joint.



   Which of the following statements about the joint is correct?

   A. 1 is the flexor.
   B. The contraction of 1 causes the relaxation of 3.
   C. 2 is elastic.
   D. 2 reduces friction between 4 and 5. ☐

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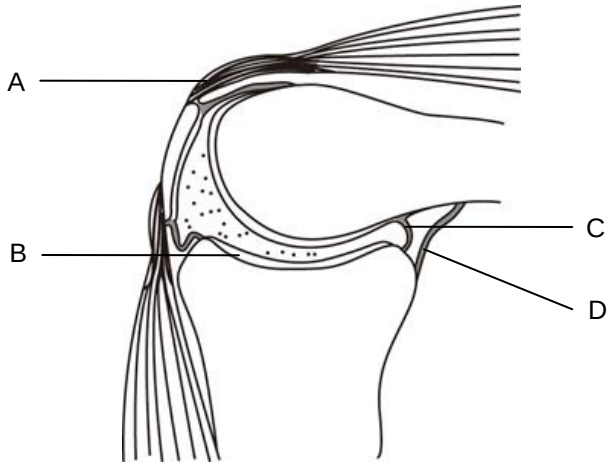
10. Which of the following statements about synovial fluid is correct?

    A. It is secreted by ligaments.
    B. It removes waste products from the surrounding tissues.
    C. It serves as a lubricant to reduce friction.
    D. It prevents the bone from becoming brittle. ☐

Question	1	2	3	4	5	6	7	8	9	10
Related section	18.1	18.1	18.1	18.2	18.1	18.2	18.3	18.5	18.4	18.2

B. Problem solving (22 marks)

11. The diagram below shows the structure of a synovial joint.



(a) Name structures B and C, and state their functions. (4 marks)

(b) Explain how the physical properties of A and D facilitate their functions respectively. (4 marks)

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(c) When a joint is damaged by injury or degeneration, it can be replaced by an artificial joint. Two artificial joints, P and Q, are shown below.



P



Q

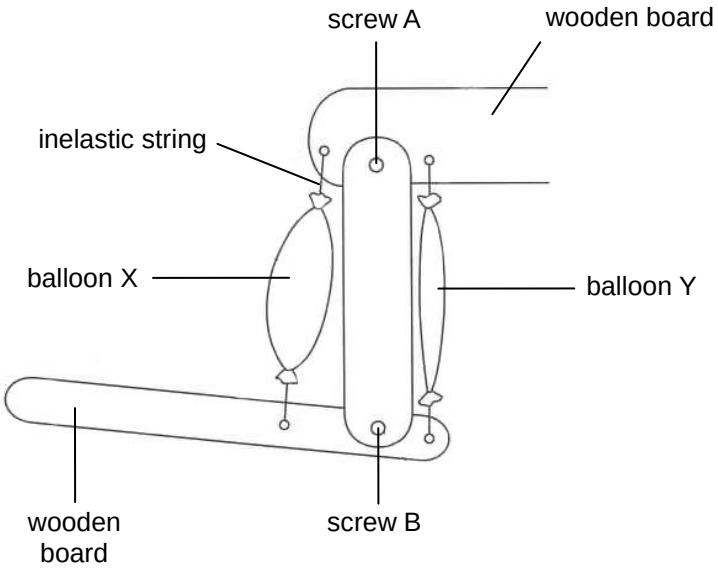
Name a joint in the human body that can be replaced by

(i) artificial joint P. (1 mark)

(ii) artificial joint Q. (1 mark)

(Total: 10 marks)

12. The diagram below shows a model used to demonstrate the movement of the human arm.



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(a) Complete the table below. (3 marks)

Component of the arm model	Corresponding structure in the human arm
wooden board	bone
screw A	shoulder joint
screw B	elbow joint
balloon X	
balloon Y	
inelastic string	

(b) Explain why the relative movement between the wooden boards at screw A cannot fully illustrate the degree of movement between the corresponding structures in the human shoulder. (2 marks)

(Total: 5 marks)

★ 13. Diagrams I and II below show a man lifting a heavy box. Diagram III shows the major muscles in his left leg.

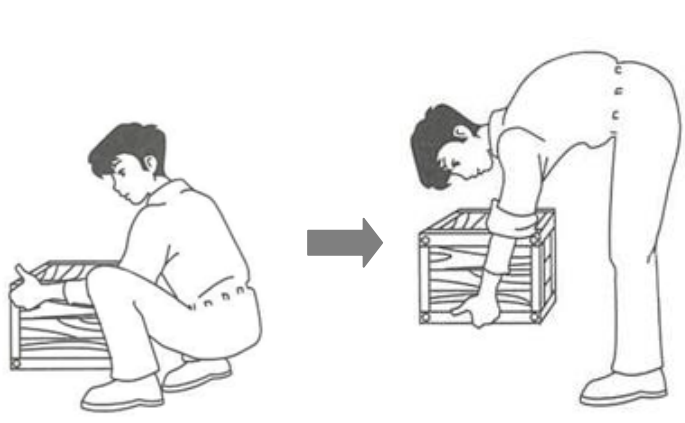


Diagram I ***Diagram II***

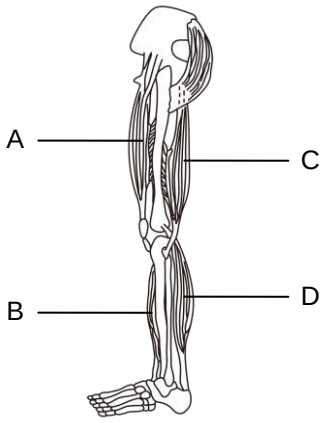


Diagram III

(a) With reference to diagram I, which muscles of the left leg (A, B, C or D) of the man is / are in a contracted state? (1 mark)

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(b) What is the term used to describe the pair of muscles A and C? Explain your answer. (2 marks)

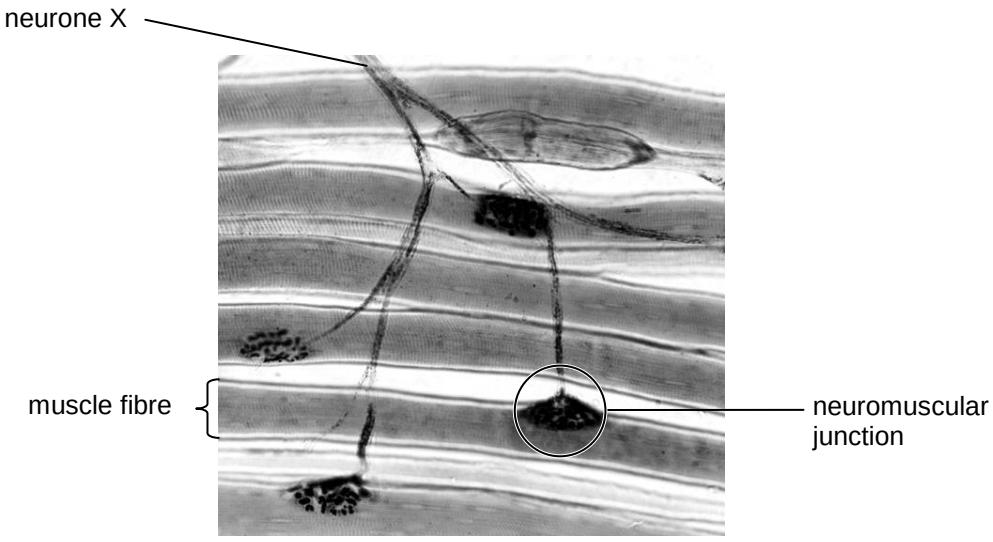
(c) Give ***two*** structural features of the vertebral column which allow it to bend as shown in diagram II (2 marks)

(d) The way the person lifts the heavy box is incorrect. Suggest an improvement in his posture for lifting the box and explain its importance. (2 marks)

(Total: 7 marks)

C. Exam challenge (8 marks)

★★ 14. The photomicrograph below shows several muscle fibres associated with the terminal parts of a neruone, X.



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(a) What type of neurones does X belong to? Explain your answer. (2 marks)

(b) With reference to the photomicrograph, explain why neurone X can cause a powerful contraction in the muscle. (2 marks)

(c) The Taiwanese banded krait (*Bungarus multicinctus*) produces venom that contains α-Bungarotoxin. The molecular shape of α-Bungarotoxin is very similar to that of acetylcholine, a neurotransmitter essential for communication between nerves and muscles at neuromuscular junctions. Suggest why paralysis will result after being bitten by the krait. (4 marks)

(Total: 8 marks)

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Marking scheme

A. Concept checking (10 marks, 1 mark each)

1.	A	2.	D	3.	C	4.	D	5.	C
6.	B	7.	A	8.	D	9.	B	10.	C

(1 mark each)

B. Problem solving (22 marks)

11. (a) B: Cartilage (1)
It acts as a shock absorber / reduces friction / prevents wearing of bone during movement. (1)
C: Synovial membrane (1)
It secretes synovial fluid to lubricate and nourish the joint. (1)
(b) A is inelastic. (1)
It can effectively transmit the pulling force produced by muscle contraction to bones. (1)
D is elastic. (1)
It holds bones in position while allowing the bones to move. (1)
(c) (i) Knee joint / Elbow joint (1)
(ii) Hip joint / Shoulder joint (1)

(Total: 10 marks)

12. (a) Balloon X: biceps (1)
Balloon Y: triceps (1)
Inelastic string: tendon (1)
(b) The shoulder joint allows the upper arm to move in many planes, (1)
but in the model the wooden board can move in one plane only. (1)

(Total: 5 marks)

13. (a) B and C (1)
(b) Antagonistic / Opposing muscles (1)
In bringing about the movement at the knee joint, muscle C contracts while muscle A relaxes. / Contraction of muscle C brings about bending of the leg while that of muscle A brings about straightening of the leg. (1)
(c) The vertebral column is made up of many vertebrae. (1)
There are compressible intervertebral discs between the vertebrae. (1)

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(d) He should keep his spine straight when lifting the box. (1)
It can reduce the bending stress exerted on the vertebral column. / It can prevent straining the vertebral column. / It can prevent dislocation of the intervertebral discs. (1)

C. Exam challenge (8 marks)

14. (a) Motor neurone (1)
It is connected to an effector (i.e. muscle fibre). (1)

(b) Neurone X forms neuromuscular junctions with several muscle fibres. (1)
It can stimulate the muscle fibres to contract simultaneously, which leads to a powerful contraction in the muscle. (1)

(c) Normally, the arrival of nerve impulses at the motor nerve ending (1)
stimulates the release of acetylcholine into the neuromuscular junction. (1)
After being bitten by the krait, α-Bungarotoxin molecules that have entered the body will compete with acetylcholine molecules for the receptor sites as they have a shape similar to acetylcholine. (1)
Therefore, less acetylcholine can bind to the receptor sites on the membrane of the muscle fibre to trigger muscle contraction. (1)

(Total: 8 marks)

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